# How the Experiments Answer the Three Questions

All three collapse into one measured quantity — the tipping point $\mathbf{p_c}$ (smallest committed/adversarial fraction that flips or breaks the system). **Lower $\mathbf{p_c}$ = more vulnerable.**

**MASTER METRIC · $p_c$( composition , scale )**

One factorial sweep — composition × model-scale × committed-fraction p — produces $p_c$, then it is read three ways. **Q3 measures $p_c$; Q1 reads how it shifts with composition; Q2 reads how it shifts with scale.**

▼ **read the same data three ways** ▼

**QUESTION 1**

## Homogeneity vs Heterogeneity

*"More vulnerable with identical vs heterogeneous LLMs?"*

**MANIPULATE**

Agent pool: **homogeneous** (all one LLM) vs **heterogeneous** (mixed families, competence-matched) vs mixed-competence.

▼

**READ FROM DATA**

**$\Delta p_c$ between compositions · correlation ρ → effective agents $N_{eff}$ · injection spread rate (GovSim)**

**Logic: one rhetoric flips *all* clones (high ρ, low $N_{eff}$) → monoculture flips at lower $p_c$. Diverse pools decorrelate → higher $p_c$.**

**QUESTION 2**

## Model Scale

*"How does model size influence robustness / susceptibility?"*

**MANIPULATE**

Model size **0.5B → 70B → frontier**, two sub-experiments: scale the **majority** (defender) and the **committed minority** (attacker) separately.

▼

**READ FROM DATA**

**$p_c$ vs size (sign = robust or fragile) · bias h vs herding J · competence covariate (scale ≠ capability)**

**Logic: majority $p_c$ ↑ = stubborn/robust; $p_c$ ↓ = sycophantic/fragile. Minority $p_c$ ↓ with size = capable attacker (dual-use).**

**QUESTION 3**

## Tipping Point

*"Is there a quantifiable threshold where cooperation degrades / vulnerability spikes?"*

**MANIPULATE**

Sweep the **committed / defector fraction p** on a fine grid; vary system size **N**.

▼

**READ FROM DATA**

**$p_c$ itself (with CI): P(consensus→A) jumps 0→1 · $T_c$ scaling exp(N) vs ln N · vs analytic 9.79%**

**Logic: the jump location = the number. Finite-size scaling confirms a true phase transition; GovSim shows it recurs in cooperation.**